

Typing Rules

Types

$$\tau ::= \text{Int} \mid \text{Bool} \mid V$$

Typing Judgements

- Expression typing:

$$\Gamma; \Sigma \vdash e : \tau$$

- Statement typing:

$$\Gamma; \Sigma \vdash s \checkmark (\checkmark \text{ indicates well-typed})$$

Literals

$$\frac{}{\Gamma; \Sigma \vdash n : \text{Int}} \text{T-Num}$$

$$\frac{}{\Gamma; \Sigma \vdash b : \text{Bool}} \text{T-Bool}$$

Variables

$$\frac{x : \tau \in \Gamma}{\Gamma; \Sigma \vdash x : \tau} \text{T-Var}$$

Binary Operations

Arithmetic

$$\frac{\Gamma; \Sigma \vdash e_1 : \text{Int} \quad \Gamma; \Sigma \vdash e_2 : \text{Int}}{\Gamma; \Sigma \vdash e_1 \oplus e_2 : \text{Int}} \text{T-ARITH} \quad (\oplus \in \{+, -, *, /\})$$

Comparison

$$\frac{\Gamma; \Sigma \vdash e_1 : \text{Int} \quad \Gamma; \Sigma \vdash e_2 : \text{Int}}{\Gamma; \Sigma \vdash e_1 \bowtie e_2 : \text{Bool}} \text{T-COMP} \quad (\bowtie \in \{<, >\})$$

Logical

$$\frac{\Gamma; \Sigma \vdash e_1 : \text{Bool} \quad \Gamma; \Sigma \vdash e_2 : \text{Bool}}{\Gamma; \Sigma \vdash e_1 \odot e_2 : \text{Bool}} \text{T-LOGIC} \quad (\odot \in \{\wedge, \vee\})$$

Equality

$$\frac{\Gamma; \Sigma \vdash e_1 : \tau \quad \Gamma; \Sigma \vdash e_2 : \tau}{\Gamma; \Sigma \vdash e_1 == e_2 : \text{Bool}} \text{T-EQ}$$

Unary Operations

$$\frac{\Gamma; \Sigma \vdash e : \text{Bool}}{\Gamma; \Sigma \vdash !e : \text{Bool}} \text{ T-NOT}$$

Variants

Variant Declaration

$$\frac{V \notin \Sigma}{\Gamma; \Sigma \vdash \text{variant } V = \{C_i(\tau_i)\} \Rightarrow \Sigma \cup \{C_i : \tau_i \rightarrow V\}} \text{ T-VARIANTDECL}$$

Variant Constructor Use

$$\frac{\Gamma; \Sigma \vdash e_j : \tau_j \quad \Sigma(C) = (\tau_1, \dots, \tau_n) \rightarrow V}{\Gamma; \Sigma \vdash C(e_1, \dots, e_n) : V} \text{ T-VARIANTCONSTR}$$

Declarations and Assignment

Variable Declaration

$$\frac{x \notin \Gamma}{\Gamma; \Sigma \vdash \text{decl } x : \tau \Rightarrow \Gamma \cup \{x : \tau\}} \text{ T-DECL}$$

Assignment

$$\frac{x : \tau \in \Gamma \quad \Gamma; \Sigma \vdash e : \tau}{\Gamma; \Sigma \vdash x := e \checkmark} \text{ T-ASSIGN}$$

Control Flow

If

$$\frac{\Gamma; \Sigma \vdash e : \text{Bool} \quad \forall s \in S. \Gamma; \Sigma \vdash s \checkmark}{\Gamma; \Sigma \vdash \text{if } e \text{ then } S \checkmark} \text{ T-IF}$$

If-Else

$$\frac{\Gamma; \Sigma \vdash e : \text{Bool} \quad \forall s \in S_1. \Gamma; \Sigma \vdash s \checkmark \quad \forall s \in S_2. \Gamma; \Sigma \vdash s \checkmark}{\Gamma; \Sigma \vdash \text{if } e \text{ then } S_1 \text{ else } S_2 \checkmark} \text{ T-IFELSE}$$

Print

$$\frac{\Gamma; \Sigma \vdash e : \tau}{\Gamma; \Sigma \vdash \text{print } e \checkmark} \text{ T-PRINT}$$

Program

$$\frac{\forall s \in P. \Gamma; \Sigma \vdash s \checkmark}{\Gamma; \Sigma \vdash P \checkmark} \text{ T-PROGRAM}$$